UNIVERSITY AGRICULTURE FAISALABAD SUB CAMPUS

(BUREWALA)

FINAL YEAR ROJECT REPORT

ON

SECURITY OPERATION CENTER DASHBOARD (SOC)

SUBMITTED TO

DEPARTMENT OF SCIENCE IN COMPUTER SCIENCE

ADVISED BY

DR. RANA MUHAMMAD SALEEM

SUBMITTED BY

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BS COMPUTER SCIENCE (8th EVENING)

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

In the Name of Allāh, the Most Gracious, the Most Merciful

i. Title: SECURITY OPERATIONS CENTER (SOC) DASHBOARD

Submitted By

1. MUHAMMAD SHOAIB-UR-REHMAN 2022-AG-8651

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Department of Computer Science

Academic Year: 2022-2026

ii. Dedication

In the name of Allah, the Most Gracious, the Most Merciful

This report is dedicated to my beloved parents,

whose unconditional love, prayers, and sacrifices

have always been my greatest strength.

To my respected teachers and mentors

who guided me in building my knowledge and skills.

iii. CERTIFICATE

This is to certify that the project entitled **"Security Operations Center (SOC) Dashboard"** submitted by **MUHAMMAD SHOAIB-UR-REHMAN & ABDUL WAHAB HAIDER,** Registration No. **2022-AG-8651 & 2022-AG-8656** in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Computer Science from **UNIVERSITY AGRICULTURE FAISALABAD SUB CAMPUS BUREWALA** is a bonafide record of work carried out by him under my supervision and guidance.

iv. ACKNOWLEDGEMENT

I take this opportunity to express my profound gratitude and deep regards to all those who contributed to the successful completion of this project. First and foremost, I am deeply thankful to my project supervisor for their exemplary guidance, valuable insights, and constant encouragement throughout the development of this work. Their constructive feedback and stimulating suggestions were instrumental in helping me coordinate and refine various aspects of this project.

Special thanks to the SOC analyst, a Cyber Security Professional, and Ethical Hacker who participated in our survey and interviews, providing crucial insights that shaped the requirements and features of this system. Their practical expertise and firsthand experience in the field were invaluable in ensuring this project addresses real-world challenges effectively.

I would like to acknowledge the unwavering support of my family, whose encouragement and understanding provided me with the strength and motivation to persevere through challenges. Their belief in my abilities has been a constant source of inspiration.

I am also grateful to my friends and colleagues who offered their assistance, shared their knowledge, and provided moral support during the course of this endeavour. Their cooperation and willingness to help made this journey more manageable and enriching.

Finally, I extend my sincere appreciation to my institution for providing the necessary resources and facilities that enabled me to undertake this project. While I have made every effort to ensure the quality of this work, I acknowledge that it would not have been possible without the collective support and guidance of everyone mentioned above.

Muhammad Shoaib-Ur-Rehman

(2022-ag-8651)

Title: SECURITY OPERATIONS CENTER (SOC) DASHBOARD

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2.Member:

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3.Member:

Mr. Waqas Ali

v. ABSTRACT

Security Operations Centers (SOCs) often struggle with scattered tools, slow threat detection, and poor incident response. This project creates a web-based SOC Dashboard that brings together security monitoring, threat visualization, and system health information in one easy-to-use platform.

The dashboard collects data from various security tools like SIEM systems, firewalls, and endpoint protection software. It uses standard web technologies React HTML, CSS, and JavaScript along with D3.js for creating visual charts, Node & Express for the backend, and MySQL for data storage. The system provides real-time updates through WebSocket technology, allowing security teams to monitor threats as they happen. By combining all essential security information in a single, user-friendly interface, this dashboard helps security professionals detect and respond to threats more quickly and effectively.

Keywords: Security Operations Center, SOC Dashboard, Threat Intelligence, Real time Monitoring, Incident Management, Cybersecurity, Data Visualization.

Table of Contents

vi. LIST OF FIGURES

vii. LIST OF TABLES

CHAPTER 1

INTRODUCTION TO THE PROBLEM

1.1 INTRODUCTION

In today's digital landscape, cybersecurity is a top priority amid surging threats, advanced attacks, and expanded risks from cloud, IoT, and remote work. Security Operations Centers (SOCs) centralized hubs for 24/7 monitoring, detection, analysis, and response rely on effective tools to manage thousands of daily alerts, yet traditional interfaces create usability hurdles like fragmented views and high false positives.

The SOC Dashboard project delivers a modern web-based platform that unifies monitoring, threat intelligence, incident management, and reporting in an intuitive interface. Built with real-time streaming, contemporary web tech, and user centered design, it tackles these issues head on. Grounded in surveys of 15 security pros and interviews with ethical hackers, this solution is timely for Pakistan's growing cybersecurity needs, where Small and Medium-sized Businesses and even established ops lack advanced visualization amid Pakistan Telecommunication Authority and National Cyber Security Centre calls for robust SOCs.

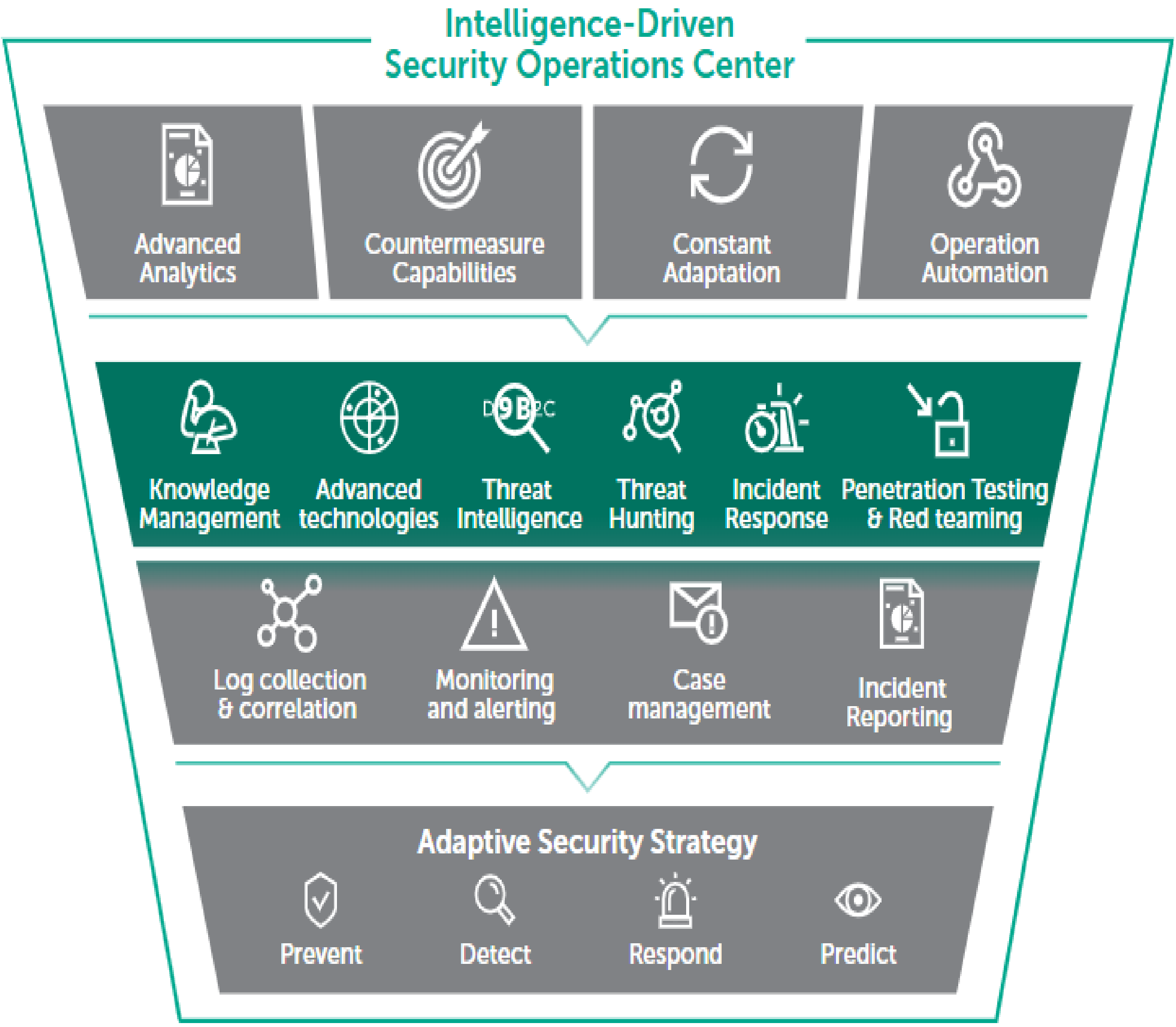
"It takes 20 years to build a reputation and few minutes of cyber-incident to ruin it." - Stephane Nappo

1.2 Problem Statement

Traditional SOC tools often present data in fragmented, non-intuitive interfaces, leading to:

- **Delayed threat detection** due to poor visualization
- **High cognitive load** on analysts from scattered dashboards
- **Inefficient incident response** caused by lack of real-time updates and centralized controls
- **Difficulty in correlating events** across multiple data sources (logs, network traffic, endpoints)

The **SOC Dashboard** solves these issues by providing a unified, real-time, and interactive platform for security monitoring, alerting, and incident management.



The foundational components of a Security Operations Center are illustrated in Figure 1.1 above.

1.3 Background

The SOC Dashboard is situated within the broader context of evolving cybersecurity operations and existing limitations in commercial and open-source SIEM platforms.

1.4 Proposed System

The proposed SOC Dashboard introduces a modern, web-based platform that overcomes legacy system limitations through intuitive design, real-time updates.

1.4.1 Key Innovations

1. **Modular Widgets:** Drag and drop grid system with 10 types (alerts, charts, maps, topology)
2. **Real-Time Streaming:** WebSockets (Socket.io) for 500ms updates on alerts, traffic, collaboration, and metrics; auto-reconnect and queuing.
3. **Advanced Visualizations:** D3.js, Chart.js, Leaflet for heatmaps, timelines, geomaps, topology diagrams, and statistical charts with interactivity.
4. **Threat Intelligence:** Integrates VirusTotal for IOC matching, TTPs (MITRE ATT&CK), and CVE context enriches alerts in real-time.
5. **Automated Response:** SOAR playbooks for remediation, tool integrations, approvals, and rollback.

1.5 Project Objectives

The SOC Dashboard targets specific, measurable goals to deliver real-time monitoring, efficient workflows, and user-friendly design.

1.5.1 Primary Objectives

(i) Real-Time Threat Visualization

- Sub-2s dashboard loads, 500ms alert delivery
- Live alerts
- Interactive geo-maps and network topology
- **Metrics:** 10+ alerts displayed

(ii) Enhanced Incident Management

- **Full lifecycle**: Create → Investigate → Resolve → Close
- Auto-creation from alerts, role-based workflows, SLA tracking
- Collaborative tools with notes and timelines
- **Metrics:** 30% faster, 90% documentation complete

(iii) Customizable User Experience

- Drag-and-drop widgets (5+ types), role-based templates
- Dark mode default, mobile-responsive
- Save/share dashboard configs
- **Metrics:** 5min customization

(iv) Intelligent Alert Processing

- Threat intel from 1+ sources (VirusTotal)
- One-click actions and custom rules
- **Metrics:** 80% alerts enriched

(v) Comprehensive Reporting

- Auto-scheduled PDF/CSV reports, 2+ templates
- Executive summaries, compliance (ISO 27001, NIST, PCI-DSS)
- Email distribution and archiving
- **Metrics:** 5s generation, 100% accuracy

Table 1.1: Secondary Objectives

Objective	Key Targets
Performance	50+ users, 1K+ events/sec, 80% uptime
Security	RBAC
Integrations	REST API, SIEM formats
Training	50+ page docs, 1hr to productivity
Scalability	50% test coverage, 15min deployments

1.6 Purpose

The purpose of this project is to develop a modern, web-based Security Operations Center Dashboard that overcomes the limitations of existing tools through intuitive design and real-time data updates. The SOC Dashboard provides a unified platform for security monitoring, threat visualization, and incident management, specifically designed to:

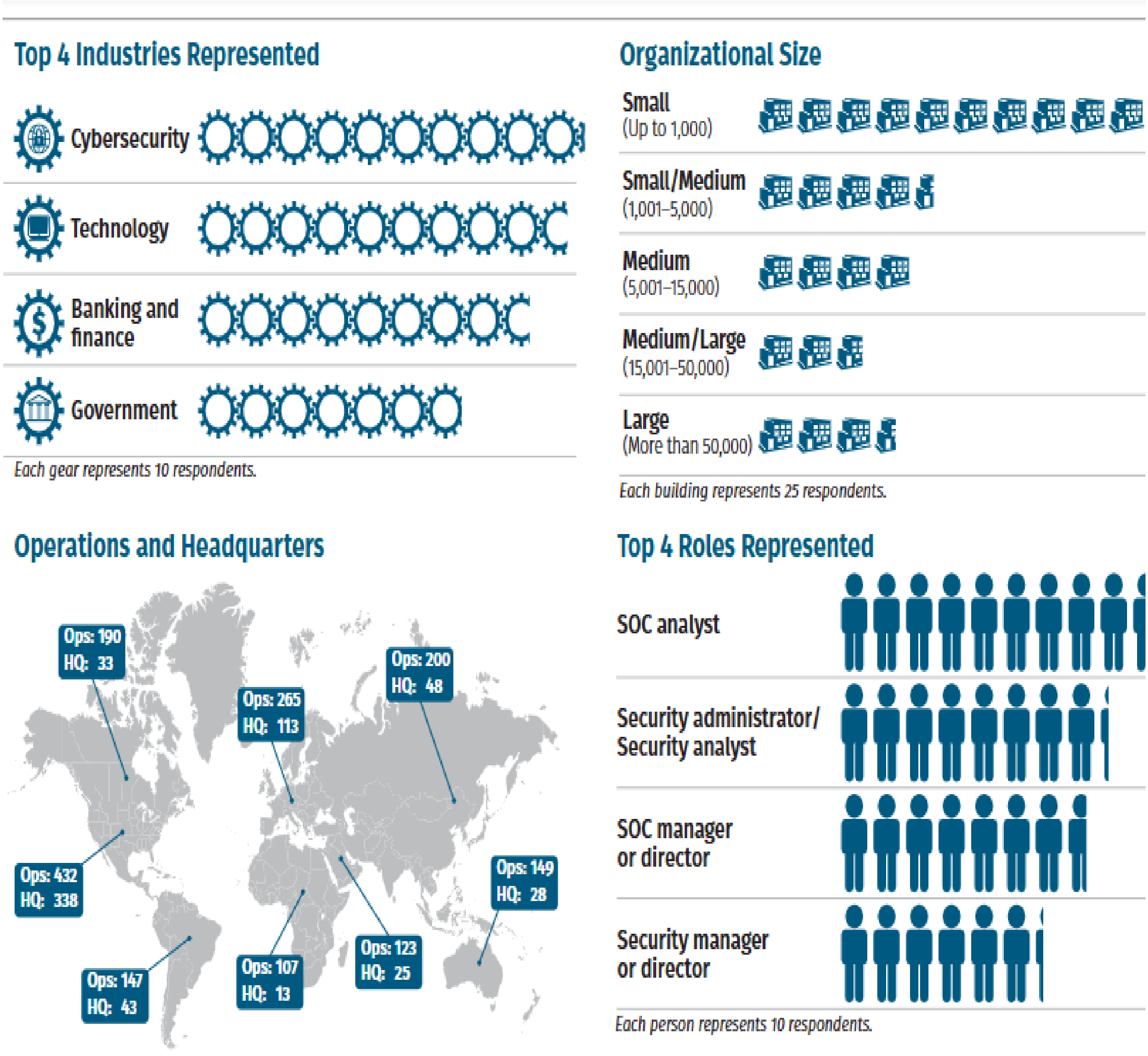
- Replace fragmented multi-tool workflows with a single, centralized interface.
- Reduce analyst cognitive load through smart visualization and automated enrichment.
- Enable faster threat detection and response through real-time streaming data.
- Provide an open-source, cost-effective alternative to expensive commercial SIEM platforms.
- Address the specific cybersecurity needs of Pakistani organizations and SMBs.

1.7 Scope

The SOC Dashboard project encompasses the design, development, and testing of a fully functional web-based security monitoring platform. The scope includes:

- A real-time security alert feed with severity-based color coding and filtering.
- Interactive data visualizations including geographic threat maps, network topology diagrams, and statistical charts.
- An incident management lifecycle from creation through investigation to resolution and closure.
- Integration with external threat intelligence services including VirusTotal, AbuseIPDB.
- Role-based access control supporting Security Analyst, SOC Manager, and System Administrator roles.
- Automated PDF and CSV report generation with scheduling capabilities.
- Performance supporting up to 5 concurrent users and many events per second.

The scope excludes deep integration with all commercial SIEM products (only REST API-compatible SIEM formats are supported) and does not include hardware procurement or physical SOC facility setup.



The demographic breakdown of the 15 security professionals surveyed is presented in Figure 1.2.

CHAPTER 2

Software Requirement Specification

2.1 Overall Description

2.1.1 Product Perspectives

The SOC Dashboard is a standalone web-based application designed to integrate with existing security infrastructure. It functions as a presentation and management layer on top of various security data sources including SIEM systems, firewalls, endpoint protection tools, and threat intelligence feeds. The system adopts a client-server architecture where the frontend renders in modern browsers while the backend processes and stores security event data.

2.1.2 Product Features

The primary features of the SOC Dashboard are summarized as follows:

- Real-time security alert monitoring with automated severity classification.
- Interactive threat visualization including geographic maps and network topology.
- Incident management with full lifecycle tracking and collaborative tools.
- Threat intelligence enrichment through integration with external feeds.
- Role-based access control with multi-factor authentication.
- Automated report generation in PDF and CSV formats.

2.1.3 Design and Implementation Constraints

- The system must function in standard web browsers (Chrome, Firefox, Edge) without requiring plugins.
- All data transmission must be encrypted via HTTPS/TLS.
- The system must comply with OWASP Top 10 security guidelines.
- Database operations must prevent SQL injection through prepared statements.

2.1.4 Assumptions and Dependencies

- It is assumed that users have access to modern web browsers with JavaScript enabled.
- The system depends on network connectivity to external threat intelligence APIs (VirusTotal, AbuseIPDB).
- MySQL 8.0 or higher is required for the database backend.
- Node.js 18.x or higher is required for the backend server.
- An active SMTP server is required for email notification functionality.

2.2 System Features

2.2.1 Real-Time Alert Monitoring

The system shall collect and display security alerts from integrated data sources in real time. Alerts shall be classified by severity (Critical, High, Medium, Low) and displayed with source IP, event type, timestamp, and recommended action. The alert feed shall update automatically via WebSocket connection without page refresh.

2.2.2 Incident Management

The system shall provide a complete incident management workflow enabling analysts to create, investigate, update, and close security incidents. Each incident shall maintain an audit trail of all actions taken, with support for collaborative notes and timeline views.

2.2.3 Threat Intelligence Integration

The system shall automatically enrich alerts with contextual threat intelligence from VirusTotal (file hash and URL reputation), AbuseIPDB (IP reputation checks. Enrichment shall be performed in near real-time for incoming alerts.

2.2.4 Automated Report Generation

The system shall generate PDF and CSV reports on demand and on a scheduled basis. Report types include incident summaries, trend analyses, executive overviews, and compliance reports. Reports shall be distributable via email and stored in an accessible archive.

2.3 External Interface Requirements

2.3.1 User Interfaces

The user interface shall be a responsive web application compatible with desktop monitors (minimum 1024x768 resolution) and mobile devices. The default theme shall be dark mode to reduce eye strain during extended monitoring sessions. Interface elements shall include color-coded severity indicators, interactive charts, filterable data tables, and a real-time alert ticker.

2.3.2 Hardware Interfaces

The system does not require any specialized hardware beyond standard server infrastructure. A minimum server configuration of 4 CPU cores, 8 GB RAM, and 50 GB storage is recommended for production deployment supporting up to 500 concurrent users.

2.3.3 Software Interfaces

The system integrates with the following external software services:

- **VirusTotal API v3:** File hash and URL reputation lookups.
- **AbuseIPDB API v2:** IP address reputation and abuse reporting.
- **SMTP servers:** Email alert distribution and report delivery.

2.3.4 Communications Interfaces

- HTTPS/TLS 1.3 for all client-server communications.
- WebSocket (Socket.io) for real-time bidirectional event streaming.
- REST API endpoints for integration with external SIEM systems.
- SMTP protocol for outbound email notifications.

2.4 Other Non-Functional Requirements

2.4.1 Performance Requirements

- Dashboard initial load time must not exceed 2 seconds on a standard broadband connection.
- The system must handle a minimum of 5 events per second without performance degradation.
- Alert delivery latency via WebSocket must not exceed 500 milliseconds.
- Report generation must complete within 5 seconds for standard report types.
- The system must support a minimum of 50 concurrent user sessions.

2.4.2 Safety Requirements

- All user actions shall be logged in an immutable audit trail for forensic purposes.
- Automatic session timeout shall occur after 15 minutes of user inactivity.
- User accounts shall be locked after 5 consecutive failed login attempts.
- All sensitive data at rest shall be encrypted using SHA-256.

2.4.3 Security Requirements

- All communications must be secured via HTTPS with TLS 1.3.
- User passwords must be hashed using bcrypt with a minimum cost factor of 12.
- RBAC (Role Based Access Control) must be supported for all user accounts.
- The system must prevent SQL injection through parameterized queries.
- Cross-Site Scripting (XSS) must be prevented through output encoding and Content Security Policy (CSP) headers.
- Role-based access control must strictly enforce the principle of least privilege.

Table 2.1: Functional Requirements Table

ID	Title	Description	Priority
FR-1	Real-Time Alert Feed	Display real-time alerts with severity badges and source information	High
FR-2	Interactive Dashboard	Provide drag-and-drop widgets for custom dashboard views	High
FR-3	Threat Intelligence Panel	Display integrated threat feeds with enrichment data	High
FR-4	Incident Management	Allow creation, update, and resolution of incident tickets	High
FR-5	Advanced Filtering	Filter alerts by IP, severity, time range, and event type	High
FR-6	Report Generation	Generate PDF/CSV reports of incidents and trends	Medium
FR-7	Role-Based Access Control	Restrict access based on user roles	High
FR-8	Live Network Map	Visualize network traffic and anomalies in real-time	Medium
FR-9	Audit Logging	Log all user actions for compliance and forensic review	Medium

CHAPTER 3

ANALYSIS (USE CASE MODEL)

3.1 Identifying Actors and Use Cases Using Textual Analysis

Textual analysis of the system requirements identifies the following primary actors and their interactions with the SOC Dashboard system.

3.1.1 System Actors

- **Security Analyst:** Monitors alerts, investigates incidents, generates reports.
- **SOC Manager:** Oversees operations, assigns incidents, reviews performance metrics.
- **System Administrator:** Manages users, configurations, and system integrations.
- **External Threat Intelligence API:** Provides automated enrichment data (VirusTotal, AbuseIPDB).
- **SIEM System:** External data source feeding security events into the dashboard.

3.1.2 Candidate Use Cases Identified from Textual Analysis

The following use cases were identified through systematic analysis of the functional requirements:

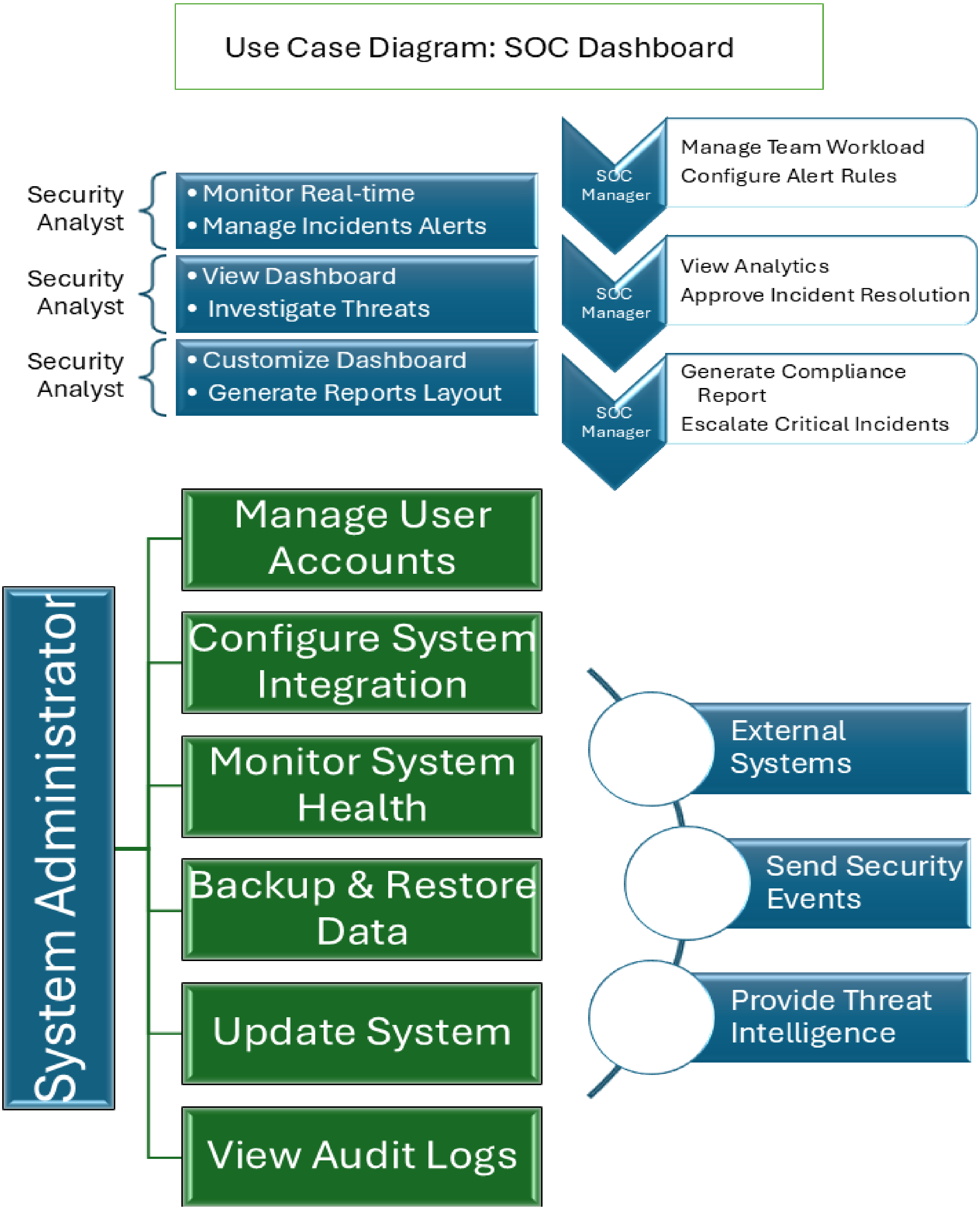
Table 3.1: Candidate Use Cases Identified Through Textual Analysis

Use Case ID	Use Case Name	Primary Actor
UC-01	User Login with RBAC	All Users
UC-02	View Real-Time Alert Dashboard	Security Analyst, SOC Manager
UC-03	Filter and Search Alerts	Security Analyst, SOC Manager
UC-04	Create Incident Ticket	Security Analyst, SOC Manager
UC-05	Investigate and Update Incident	Security Analyst
UC-06	Close/Resolve Incident	Security Analyst, SOC Manager
UC-07	Assign Incident to Analyst	SOC Manager

Use Case ID	Use Case Name	Primary Actor
UC-08	View Threat Intelligence Enrichment	Security Analyst, SOC Manager
UC-09	Generate Security Report	Security Analyst, SOC Manager
UC-10	Schedule Automated Reports	SOC Manager
UC-11	Manage User Accounts	System Administrator
UC-12	Configure System Integrations	System Administrator
UC-13	View Audit Logs	System Administrator, SOC Manager
UC-14	Monitor System Health Status	System Administrator

3.2 Forming Use Case Diagram with Candidate and Use Cases

The use case diagram below illustrates the relationships between all identified actors and use cases. The diagram shows three primary actor roles with their respective use case interactions, including extend and include relationships for complex workflows.



The complete use case diagram showing all actors and their interactions is presented in Figure 3.1 below.

The Use Case Diagram demonstrates that the Security Analyst interacts with the core operational use cases (UC-02 through UC-10), while the SOC Manager has broader access including team management functions. The System Administrator operates independently on infrastructure use cases (UC-12 through UC-15). The Login use case (UC-01) is a prerequisite for all other use cases through an include relationship.

3.3 Description of Event Flows for Use Cases

3.3.1 Use Case: View Real-Time Alert Dashboard (UC-02)

- **Primary Actor:** Security Analyst / SOC Manager
- **Precondition:** User is authenticated and has dashboard access permissions.
- **Main Flow:**
    1) User logs in and navigates to the main dashboard.
    2) System connects WebSocket and begins streaming live alert data.
    3) Dashboard widgets render real-time charts and alert ticker.
    4) User can click any alert to view detailed information.
    5) System continuously updates without page refresh.
- **Alternate Flow:** If WebSocket connection fails, system falls back to polling every 10 seconds and displays a connection status warning.
- **Postcondition:** User has a current, up-to-date view of the security posture.

3.3.2 Use Case: Create Incident Ticket (UC-04)

- **Primary Actor:** Security Analyst / SOC Manager
- **Precondition:** User is authenticated. A relevant security alert exists in the system.
- **Main Flow:**
    1) Analyst identifies a suspicious alert requiring investigation.
    2) Analyst clicks 'Create Incident' from the alert detail view.
    3) System pre-populates the incident form with alert data.
    4) Analyst reviews data, adds description and initial notes.

5) Analyst selects severity level and assigns to self or a team member.

6) System creates incident record, assigns an ID, and logs the action.

7) Assigned analyst receives a notification.

- **Alternate Flow:** If the analyst lacks assignment permissions, only self-assignment is permitted.
- **Postcondition:** A new incident record is created and visible in the incident queue.

3.3.3 Use Case: Generate Security Report (UC-10)

- **Primary Actor:** Security Analyst / SOC Manager
- **Precondition:** User is authenticated. Relevant incident and alert data exist for the requested period.
- **Main Flow:**

1) User navigates to the Reports module.

2) User selects report type (incident summary, trend analysis, compliance).

3) User sets date range and output format (PDF or CSV).

4) System retrieves and processes the relevant data.

5) System generates the report within 5 seconds.

6) Report is presented for download and optionally emailed to specified recipients.

- **Alternate Flow:** If no data exists for the selected period, the system notifies the user and offers to extend the date range.
- **Postcondition:** Report is generated, archived, and available for download.

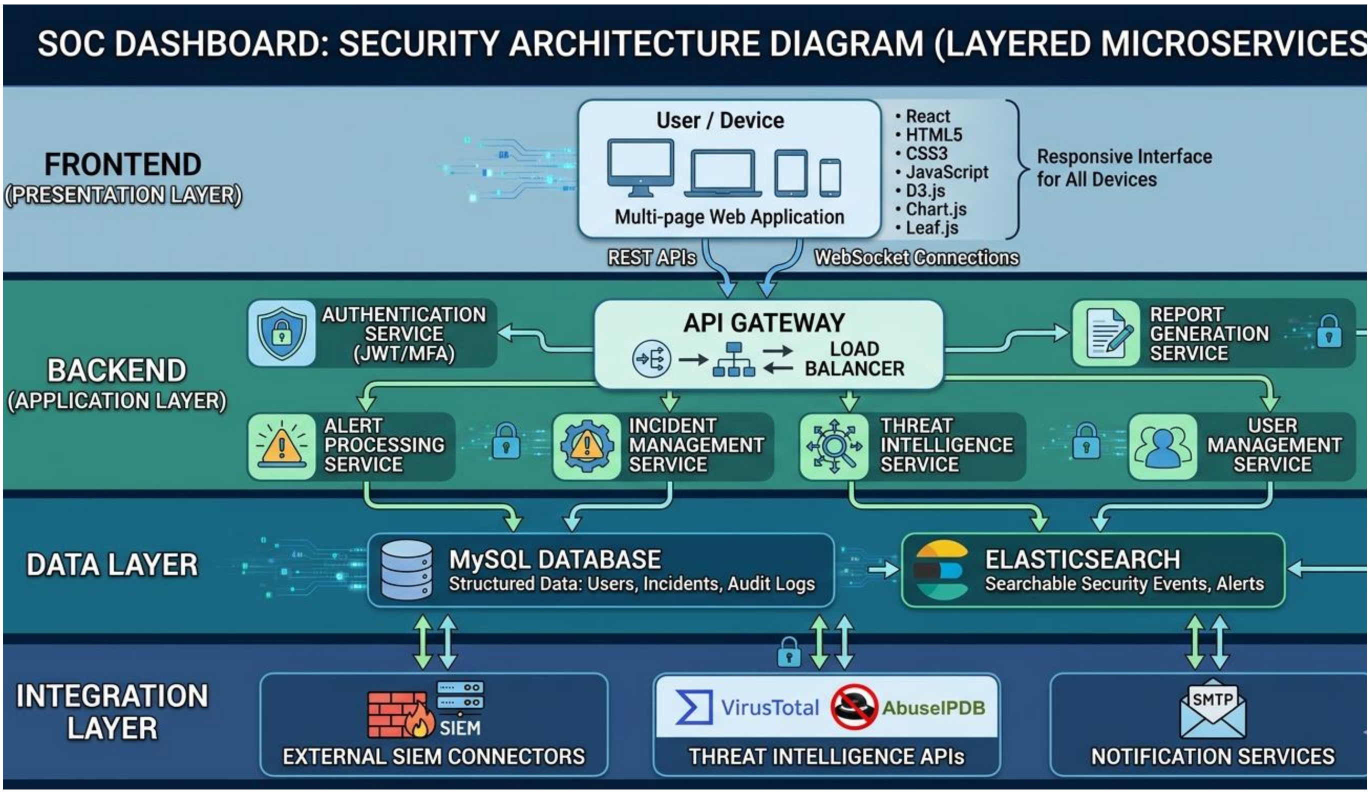
CHAPTER 4

DESIGN

4.1 Architecture Diagram

The SOC Dashboard adopts a layered microservices architecture designed for scalability, maintainability, and resilience. The architecture is organized into four principal layers:

- **Frontend (Presentation Layer):** A multi-page web application built with React , HTML5, CSS3, JavaScript, D3.js, and Chart.js or Leaf.js. The frontend communicates with the backend via REST APIs and WebSocket connections, providing a responsive interface for all device types.
- **Backend (Application Layer):** Implemented in Node.js/Express.js. An API Gateway handles all incoming requests and provides load balancing. Key microservices include: Authentication Service (JWT/MFA), Alert Processing Service, Incident Management Service, Threat Intelligence Service, Report Generation Service, and User Management Service.
- **Data Layer:** MySQL database for structured data (users, incidents, audit logs). Elasticsearch for searchable security events and alerts.
- **Integration Layer:** External SIEM connectors, threat intelligence APIs (VirusTotal, AbuseIPDB), and notification services (SMTP).



The system follows a layered microservices architecture as depicted in Figure 4.1.

4.2 Data Flow Diagrams

4.2.1 Level 0 DFD (Context Diagram)

The Level 0 DFD provides a high-level overview of data flows between the SOC Dashboard system and its external entities. External entities include Security Analysts/SOC Managers, System Administrators, external SIEM systems, threat intelligence APIs, and notification recipients.

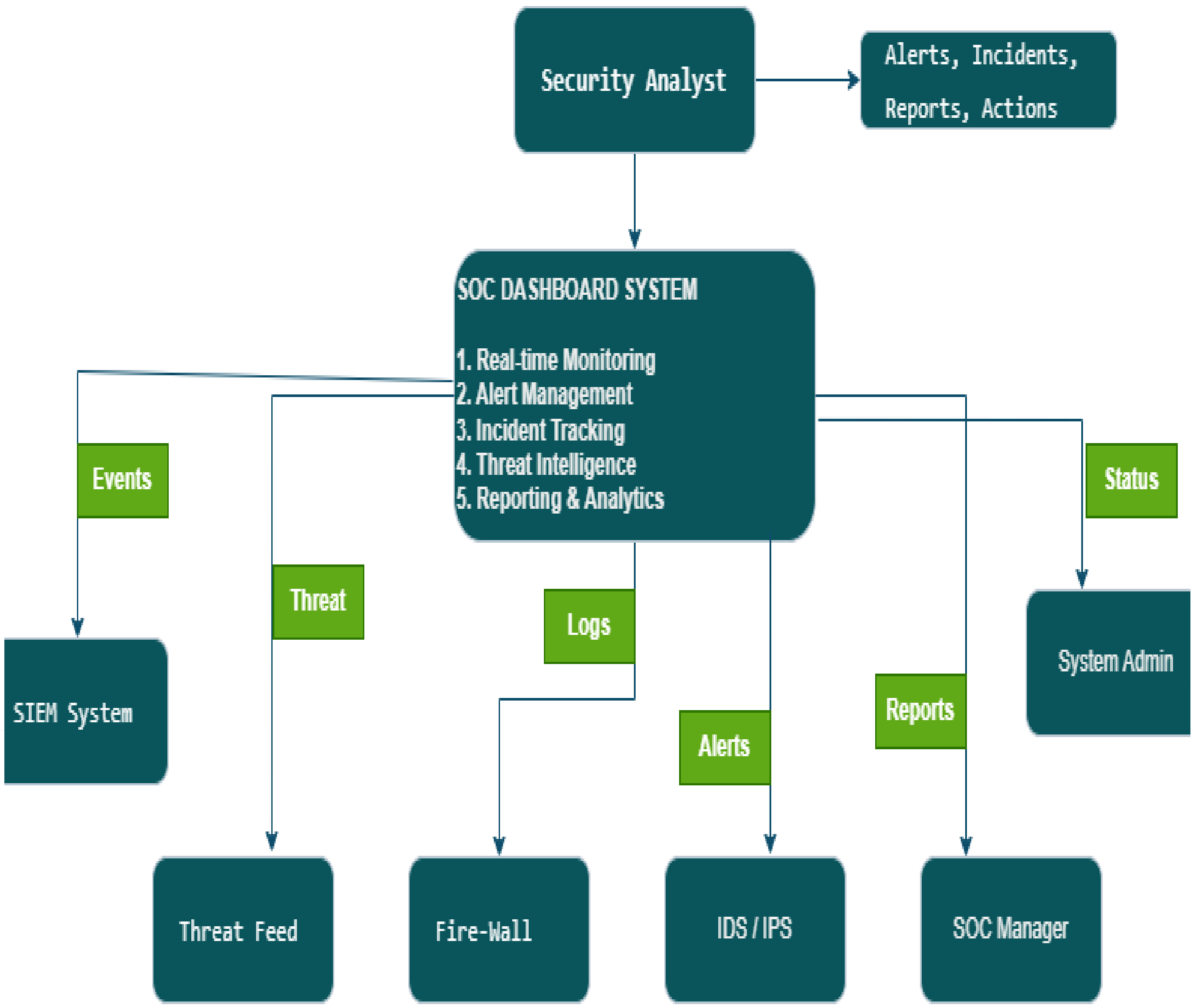
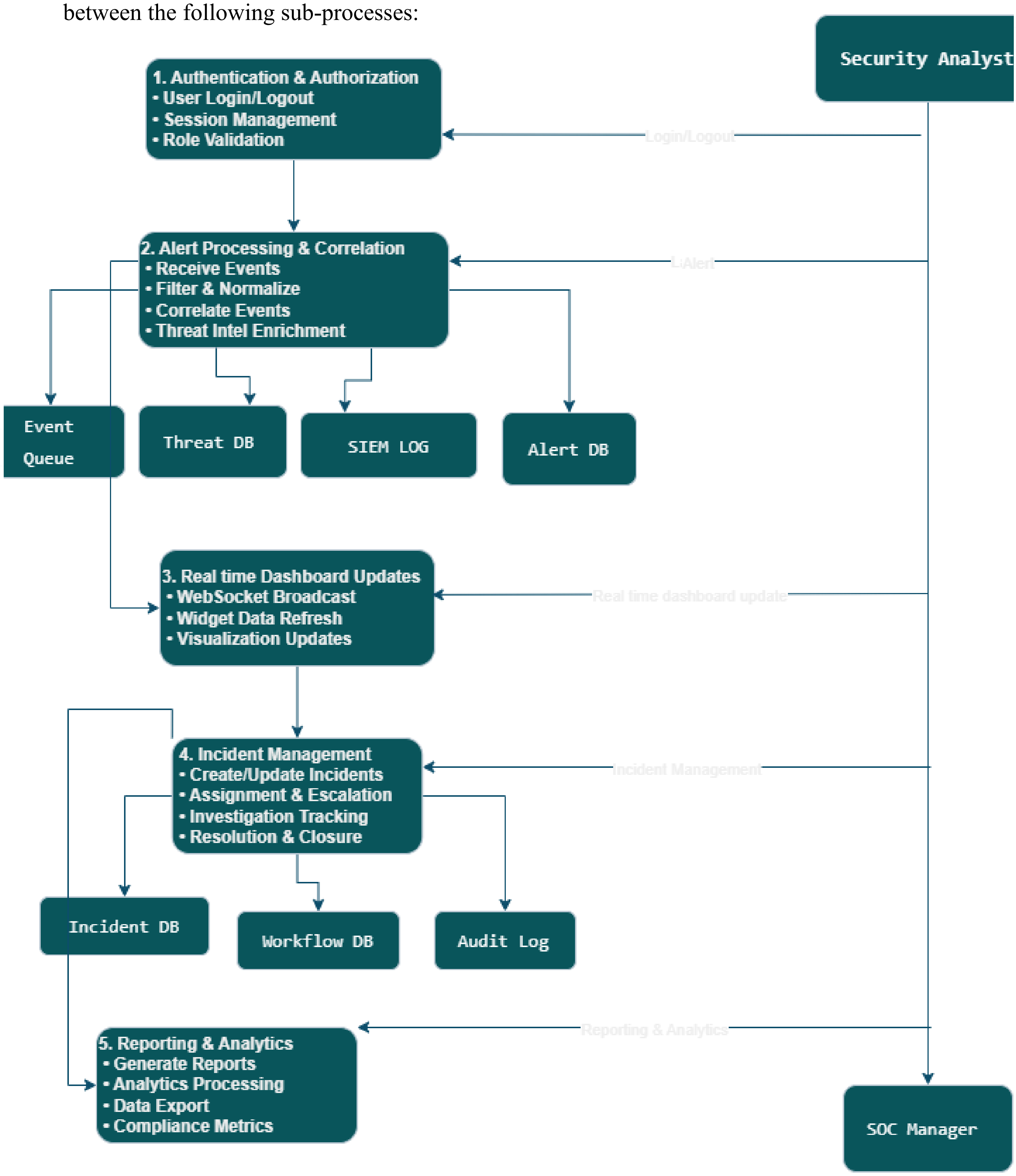


Figure 4.2.1 presents the context-level data flow diagram showing external entities interacting with the system.

4.2.2 Level 1 DFD

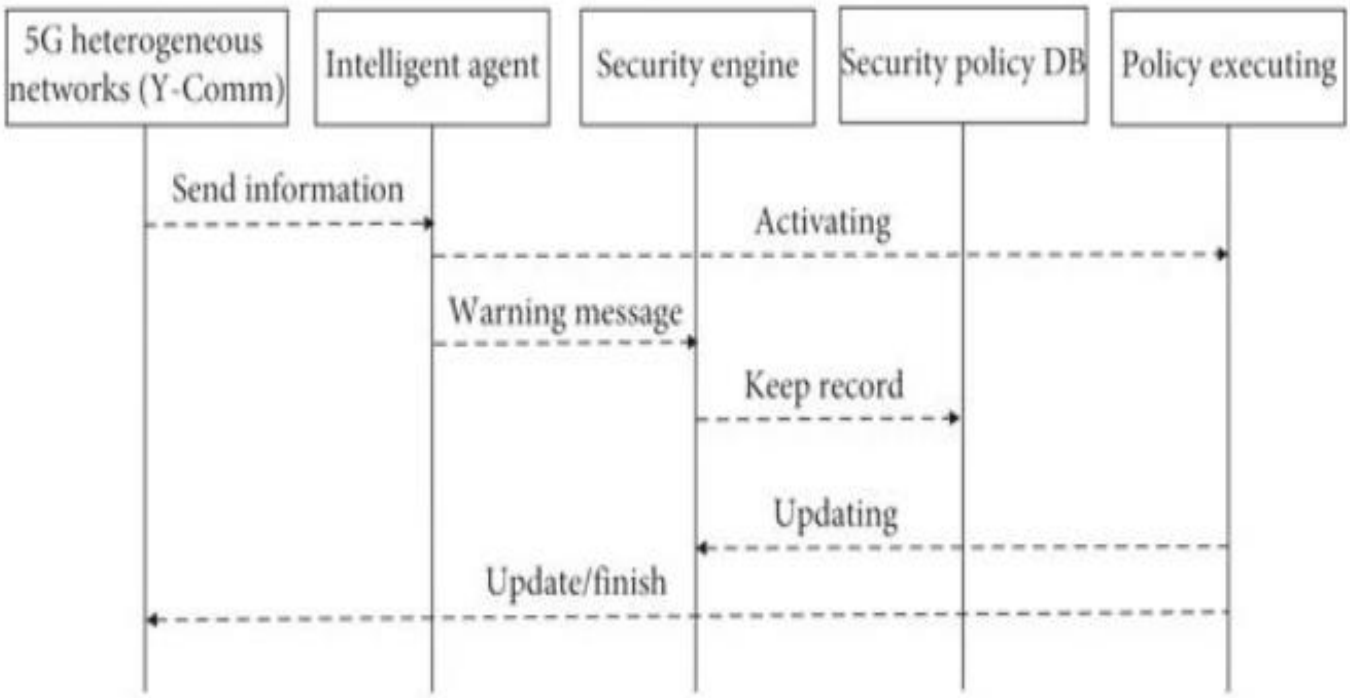
The Level 1 DFD expands the core processes of the SOC Dashboard, showing data flows between the following sub-processes:



The expanded Level 1 Data Flow Diagram is shown in Figure 4.2.2, detailing the core system processes.

4.3 Sequence Diagram

The sequence diagram below illustrates the interaction flow for real-time alert processing, from data ingestion through to analyst notification. This captures the temporal ordering of messages between system components.



The interaction flow for real-time alert processing is illustrated in the sequence diagram (Figure 4.3).

4.4 Activity Diagram

The activity diagram below represents the workflow for incident management, capturing the state transitions and decision points from initial alert detection through incident closure.

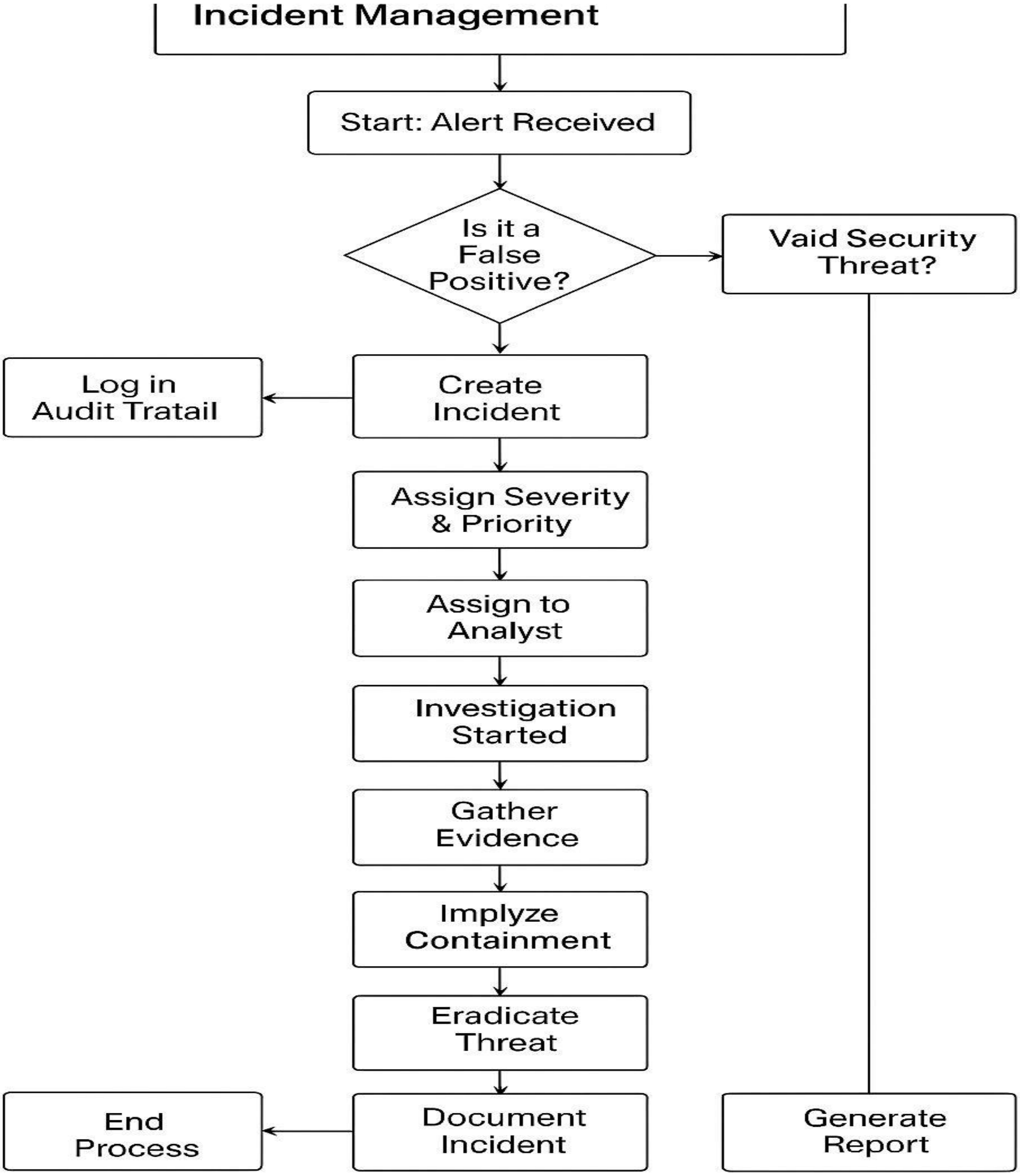
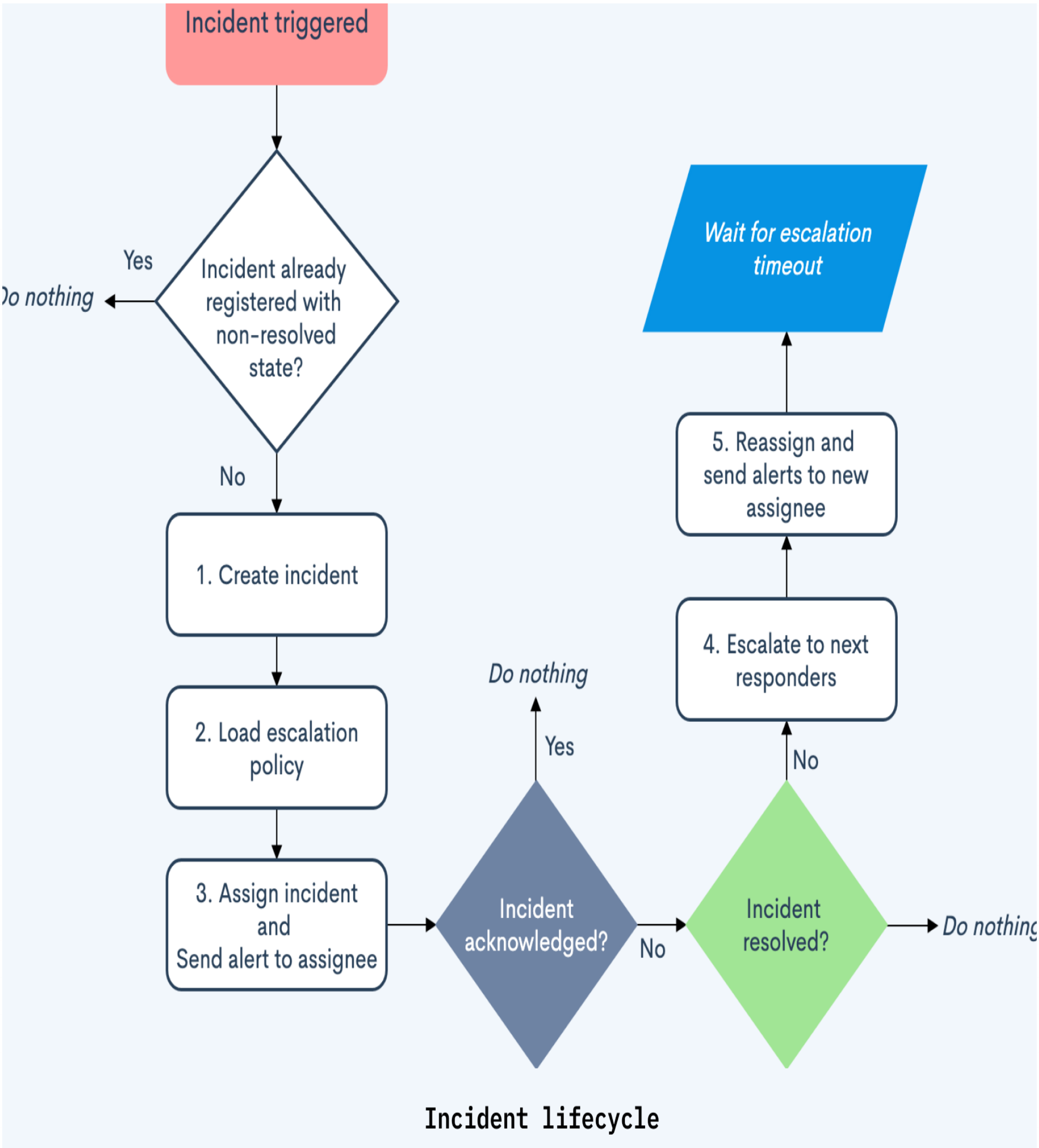


Figure 4.4 presents the activity diagram for incident management, capturing decision points and parallel flows.

4.5 State Transition Diagram

The state transition diagram models the lifecycle states of a security incident within the SOC Dashboard. The diagram captures all valid transitions and triggering events.



The state transition diagram (Figure 4.5) models the complete lifecycle of a security incident.

4.6 ERD with Data Dictionary

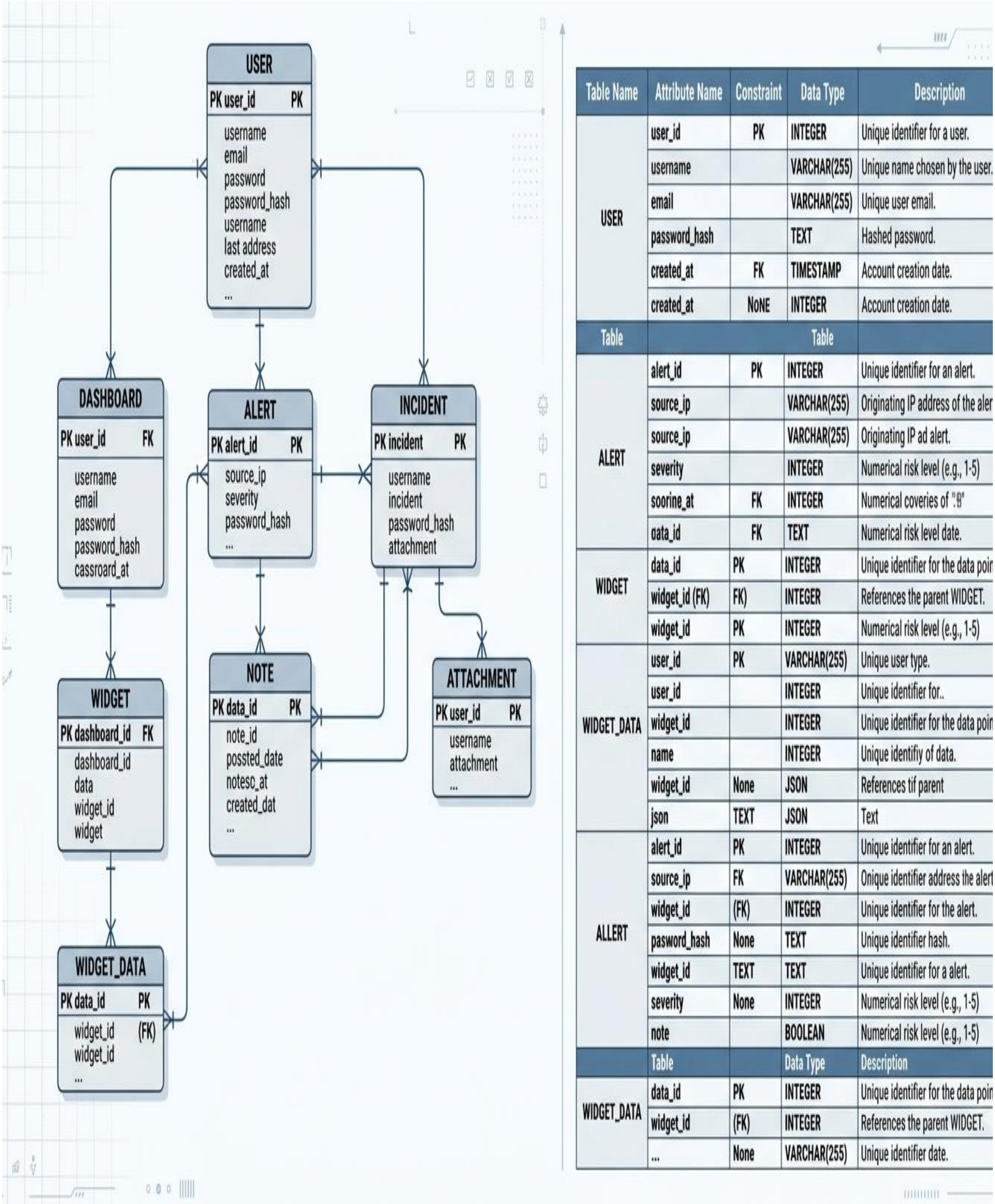


Table Name	Attribute Name	Constraint	Data Type	Description
USER	user_id	PK	INTEGER	Unique identifier for a user.
	username		VARCHAR(255)	Unique name chosen by the user.
	email		VARCHAR(255)	Unique user email.
	password_hash		TEXT	Hashed password.
	created_at	FK	TIMESTAMP	Account creation date.
	created_at	NoNE	INTEGER	Account creation date.
Table		Table		
ALERT	alert_id	PK	INTEGER	Unique identifier for an alert.
	source_ip		VARCHAR(255)	Originating IP address of the aler
	source_ip		VARCHAR(255)	Originating IP ad alert.
	severity		INTEGER	Numerical risk level (e.g., 1-5)
	soorine_at	FK	INTEGER	Numerical coveries of ".8"
	oata_id	FK	TEXT	Numerical risk level date.
WIDGET	data_id	PK	INTEGER	Unique identifier for the data poir
	widget_id (FK)	FK)	INTEGER	References the parent WIDGET.
	widget_id	PK	INTEGER	Numerical risk level (e.g., 1-5)
WIDGET_DATA	user_id	PK	VARCHAR(255)	Unique user type.
	user_id		INTEGER	Unique identifier for..
	widget_id		INTEGER	Unique identifier for the data poir
	name		INTEGER	Unique identifiy of data.
	widget_id	None	JSON	References tif parent
	json	TEXT	JSON	Text
ALLERT	alert_id	PK	INTEGER	Unique identifier for an alert.
	source_ip	FK	VARCHAR(255)	Onique identifier address the alert
	widget_id	(FK)	INTEGER	Unique identifier for the alert.
	pasword_hash	None	TEXT	Unique identifier hash.
	widget_id	TEXT	TEXT	Unique identifier for a alert.
	severity	None	INTEGER	Numerical risk level (e.g., 1-5)
	note		BOOLEAN	Numerical risk level (e.g., 1-5)
	Table		Data Type	Description
WIDGET_DATA	data_id	PK	INTEGER	Unique identifier for the data poir
	widget_id	(FK)	INTEGER	References the parent WIDGET.
	...	None	VARCHAR(255)	Unique identifier date.

The database schema is represented through the Entity Relationship Diagram in Figure 4.6.

4.7 Class Diagram

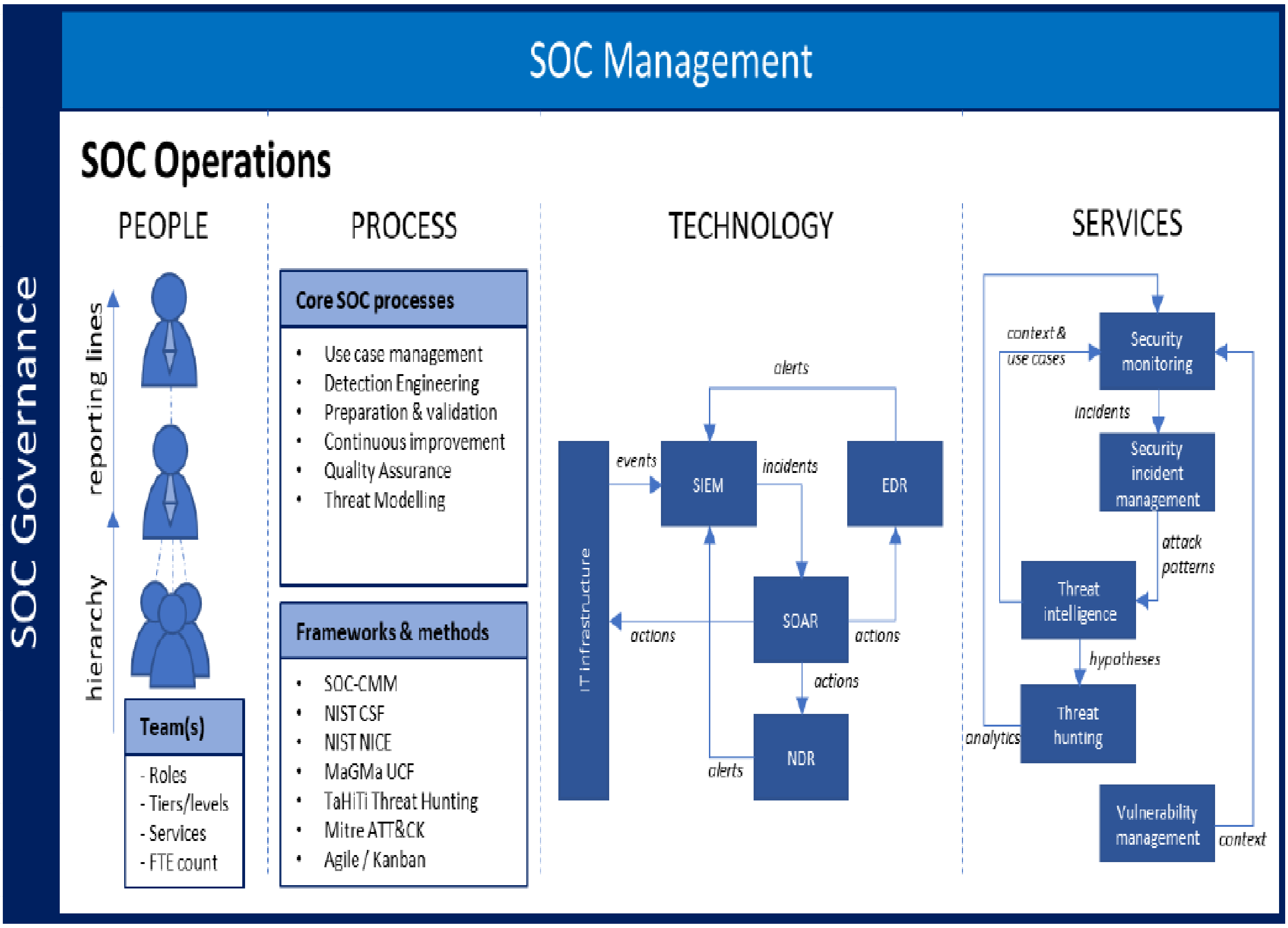
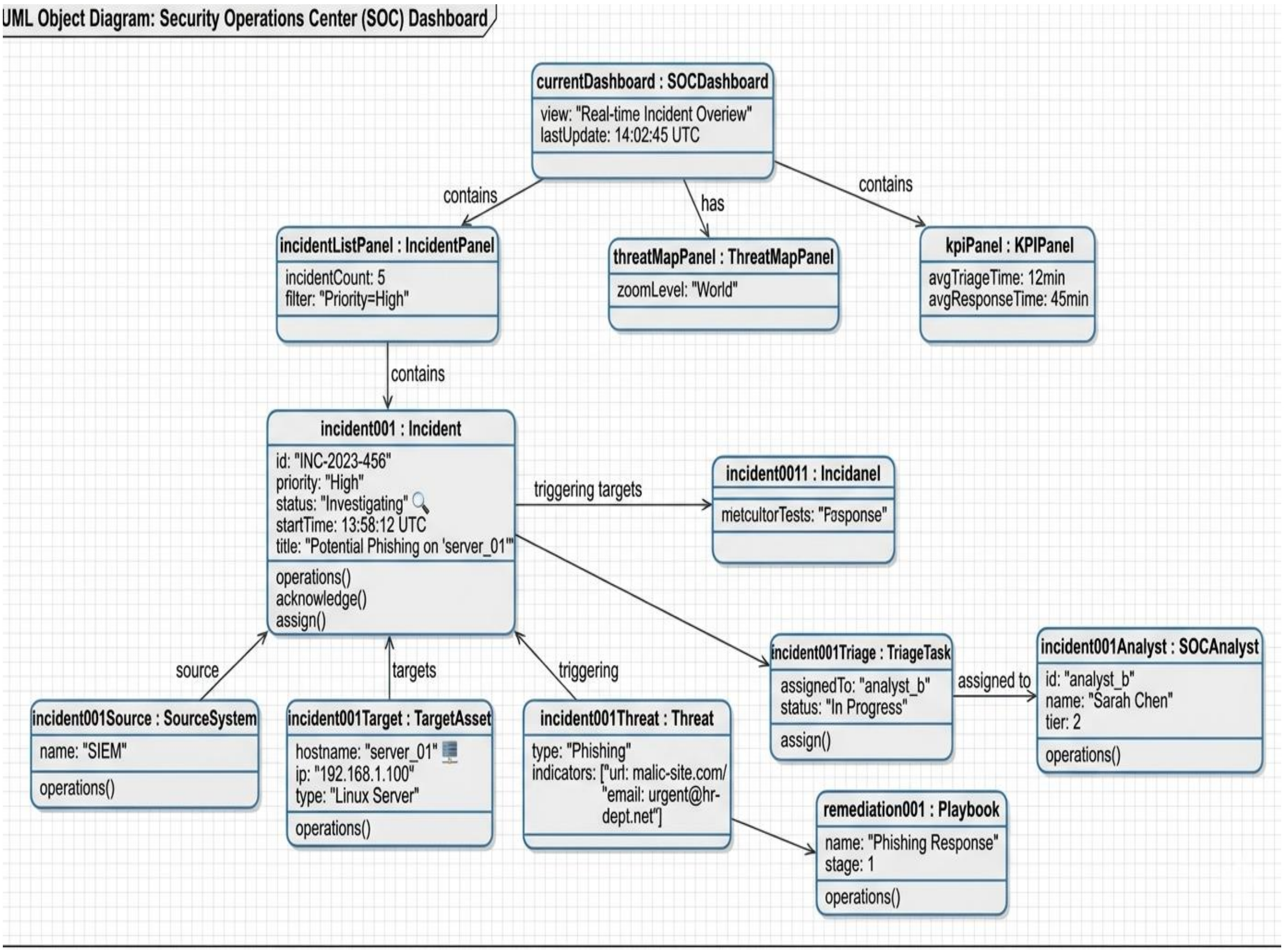


Figure 4.7 shows the class diagram representing the object-oriented structure of the SOC Dashboard.

4.8 Object Diagram



A concrete instance of the class diagram is illustrated in the object diagram (Figure 4.8).

4.9 Collaboration Diagram

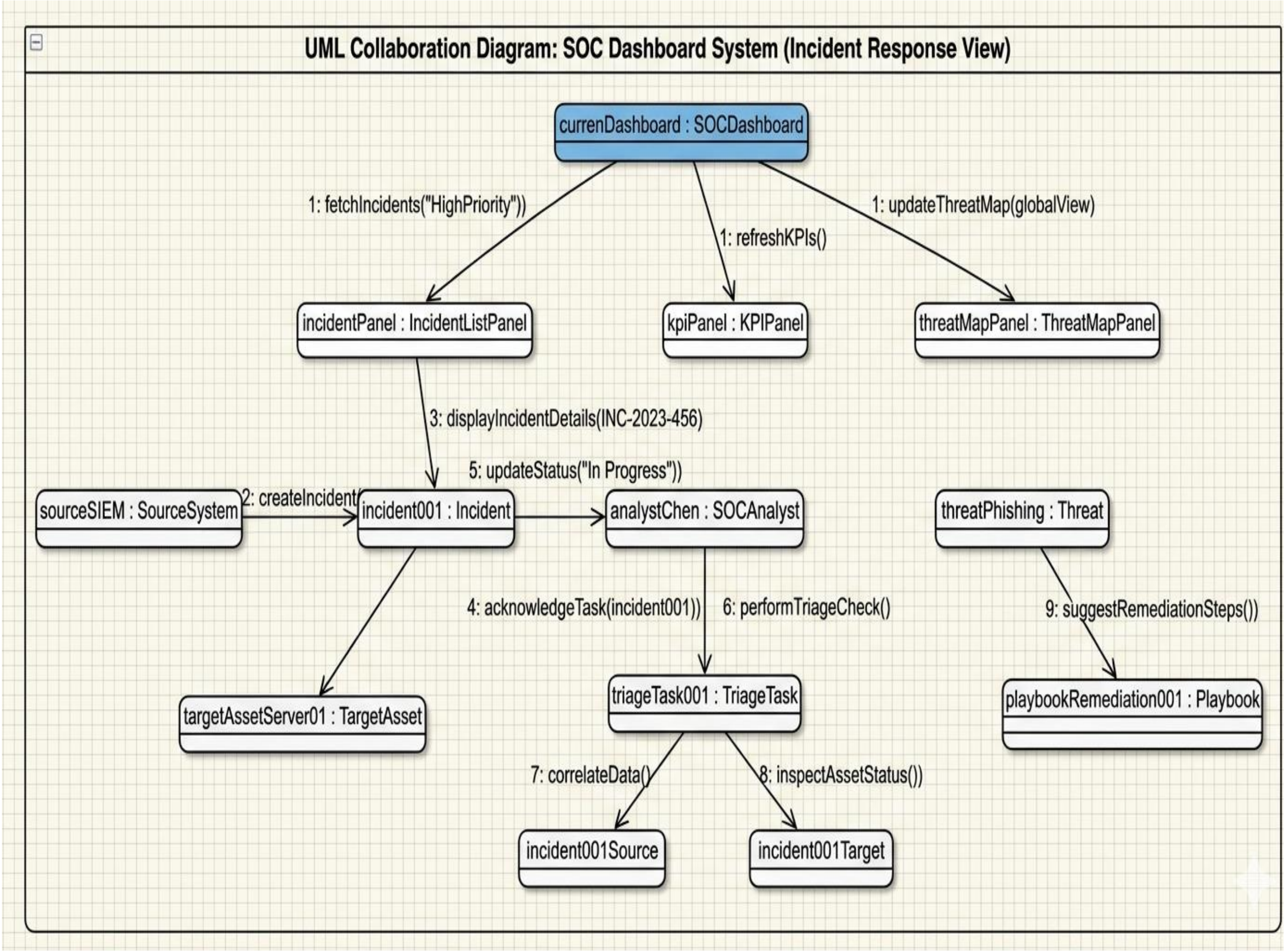


Figure 4.9 presents the collaboration diagram showing message passing between objects.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 Dashboard Interfaces

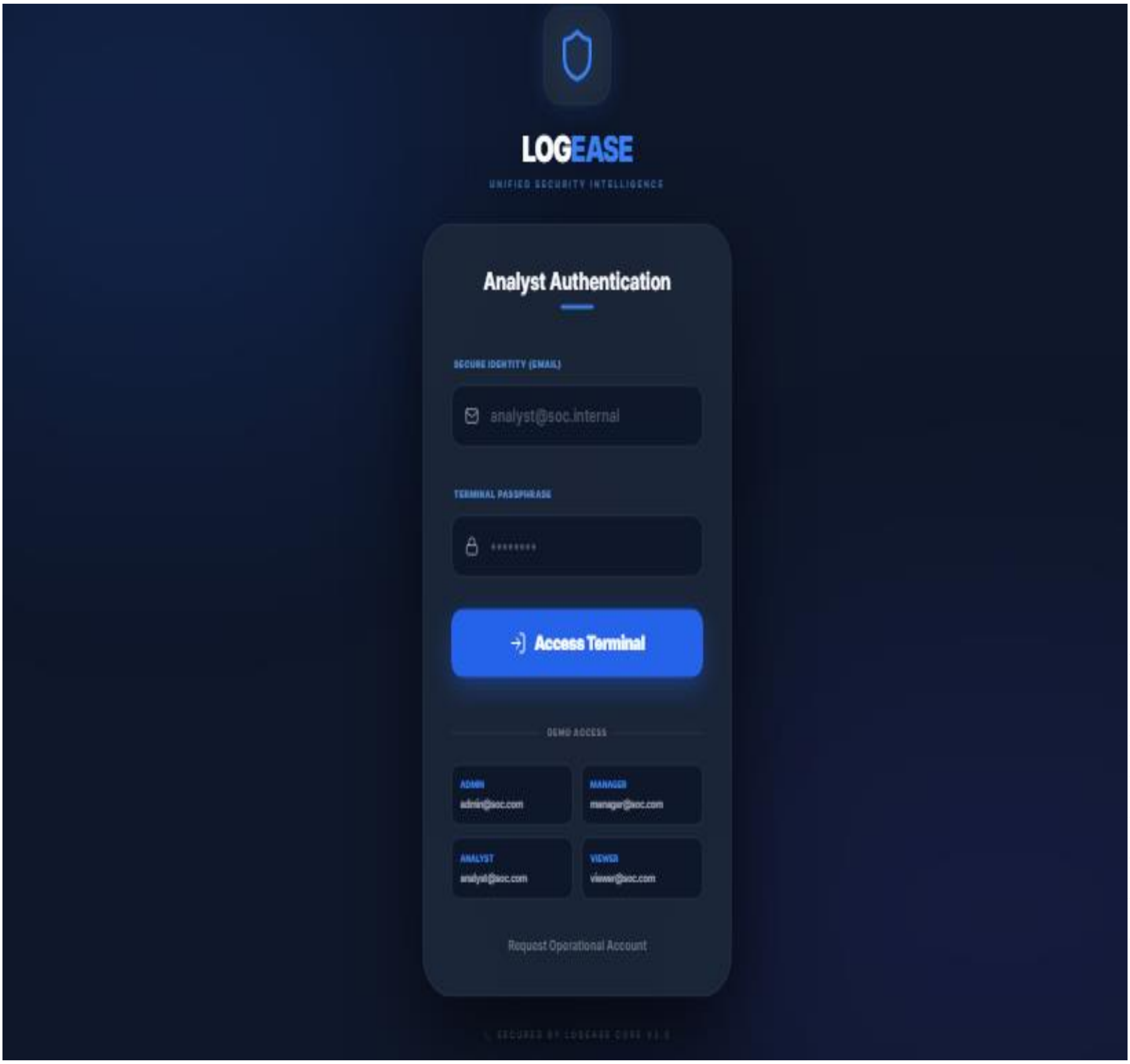
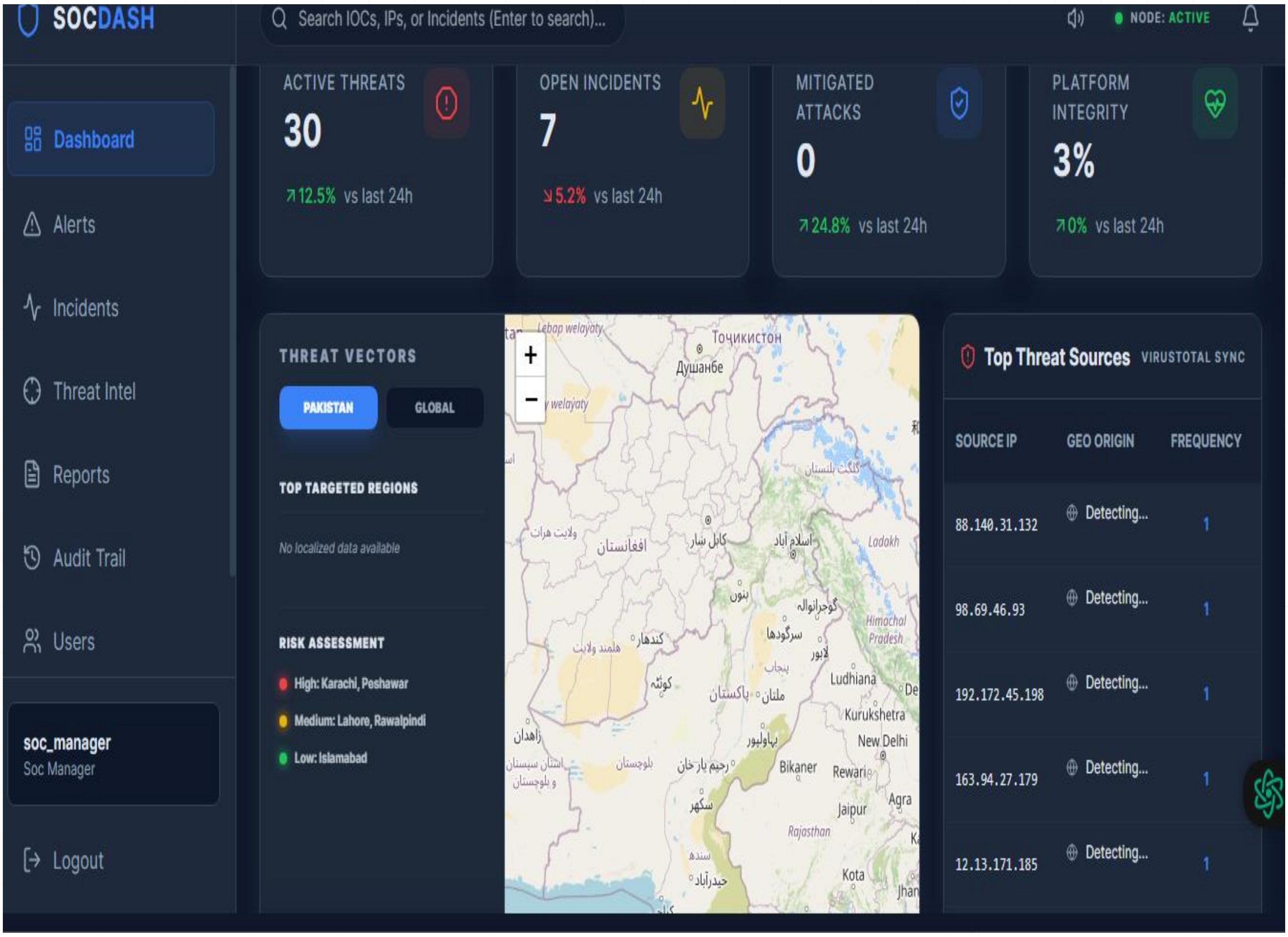


Figure 5.1 displays the secure login interface of LOGEASE.



The main dashboard view with real-time metrics is shown in Figure 5.2.

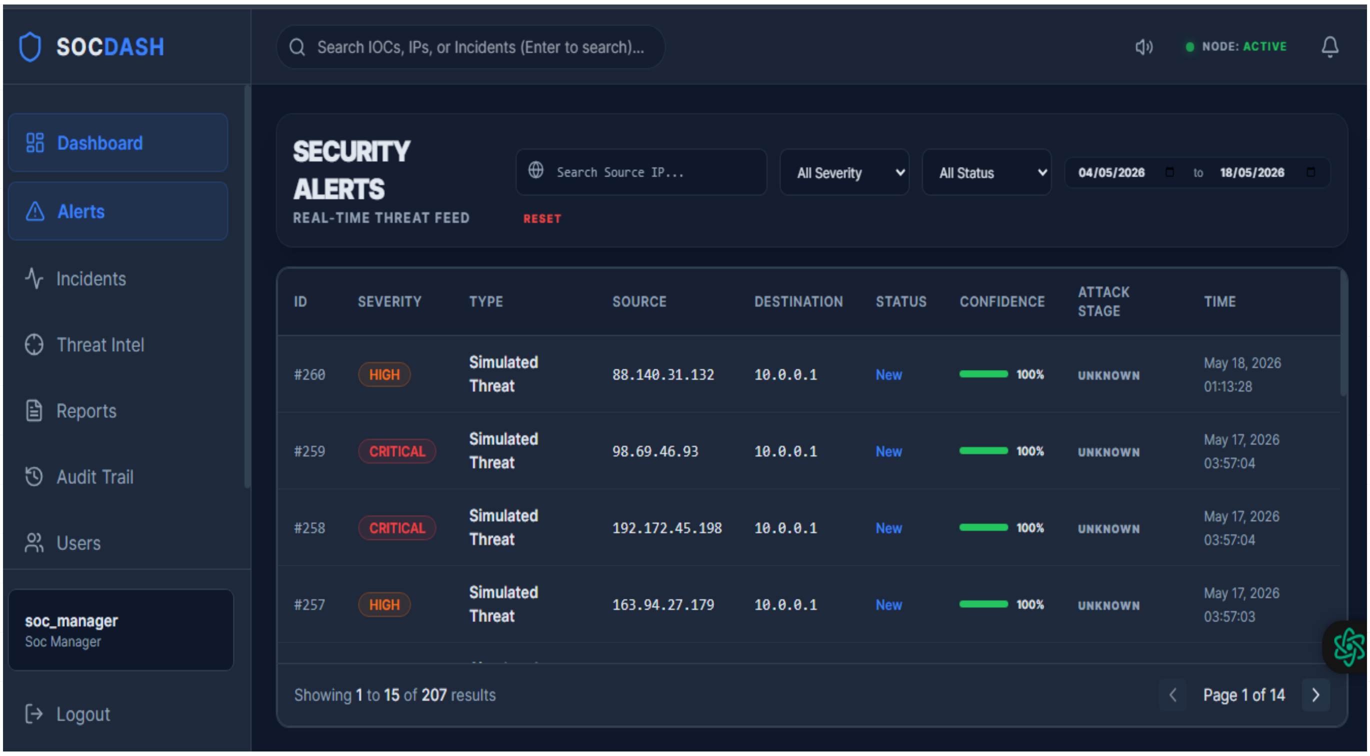
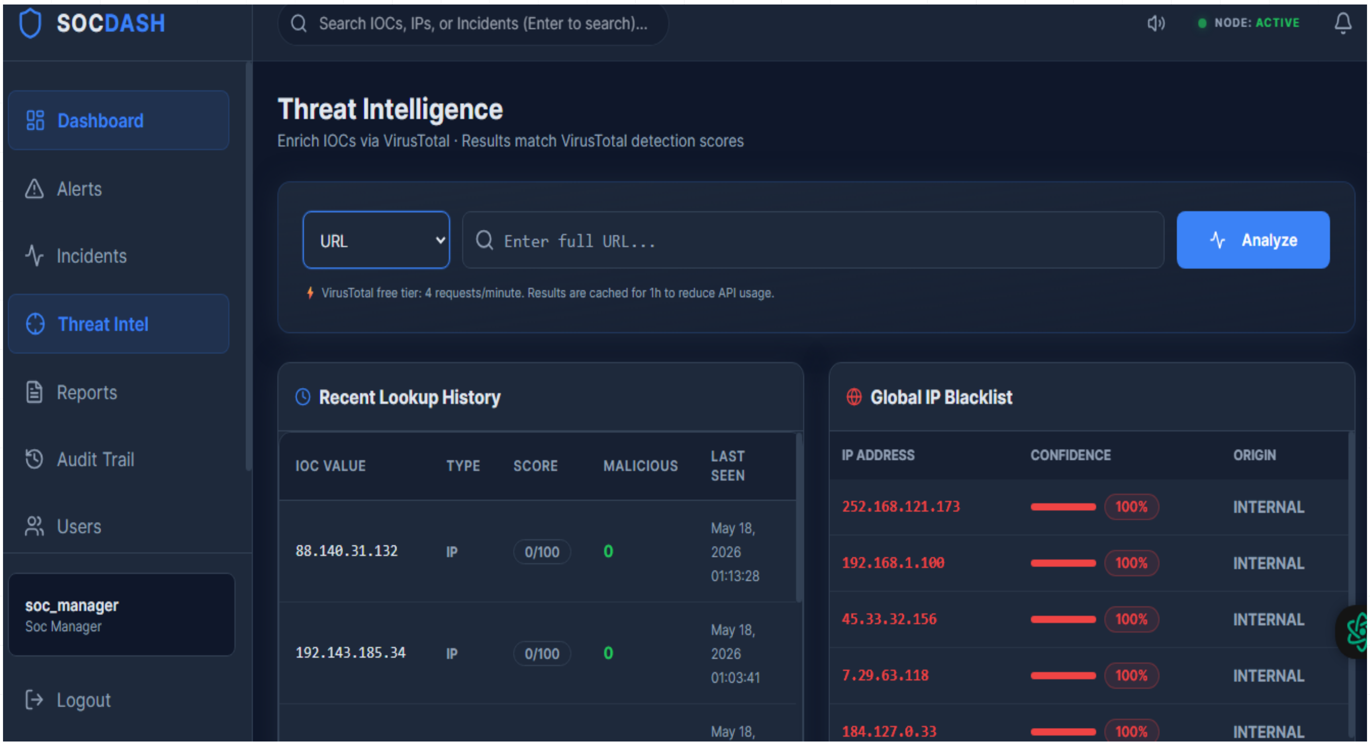


Figure 5.3 presents the alert monitoring page with severity-based color coding.



The threat intelligence integration page is illustrated in Figure 5.4.

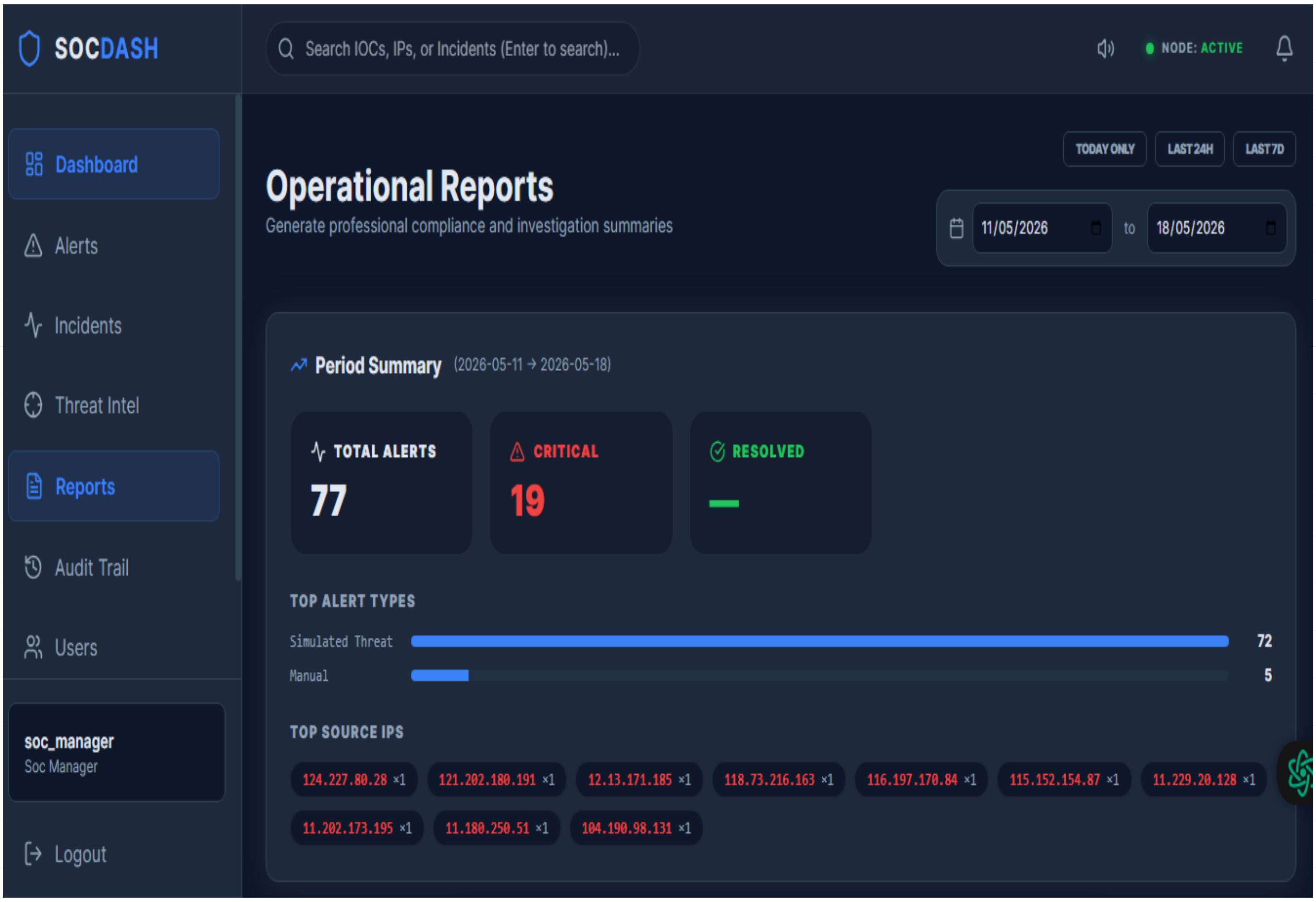
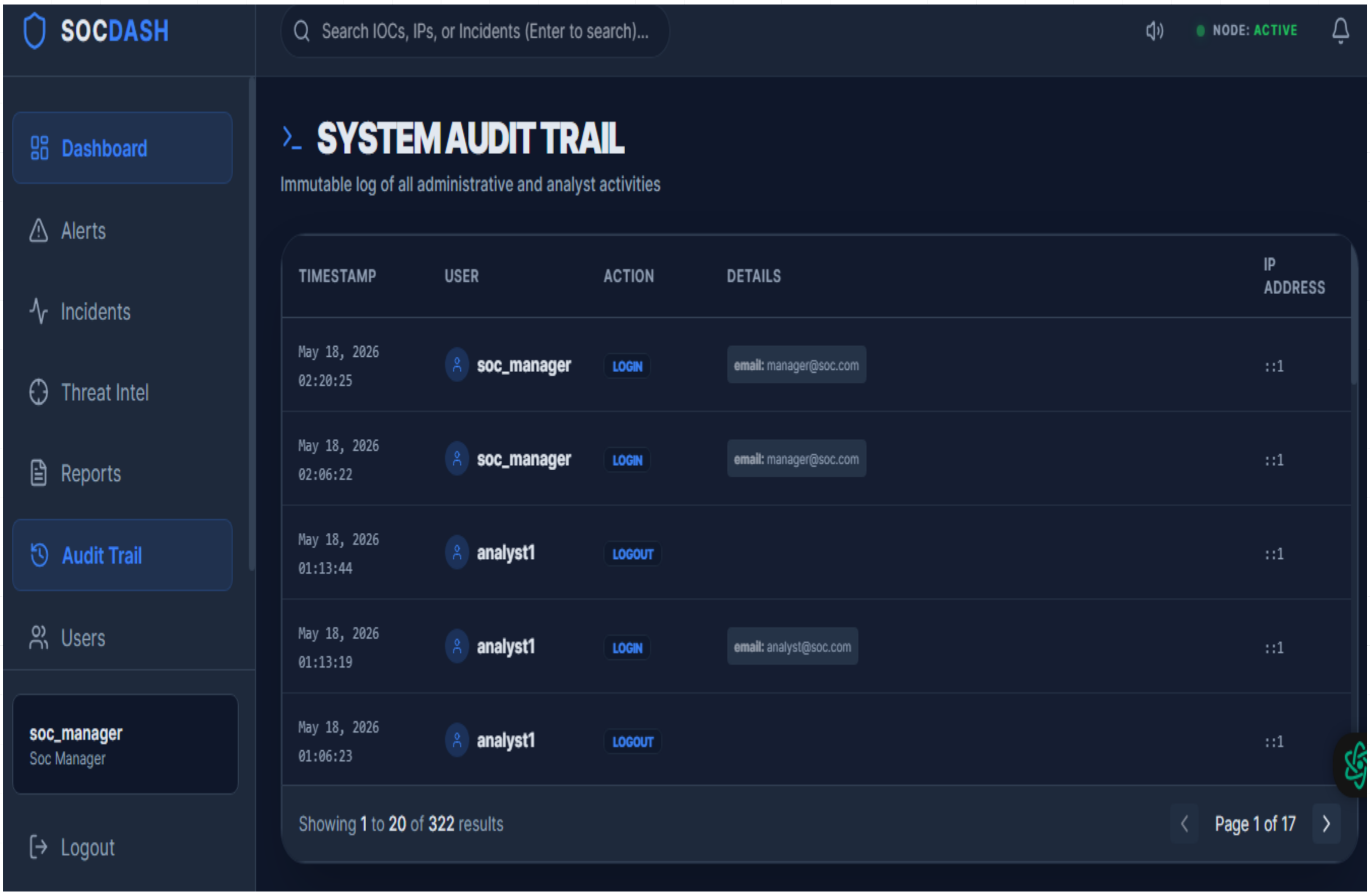


Figure 5.5 shows the reporting page with PDF/CSV generation options.



The audit trail page for compliance tracking is presented in Figure 5.6.

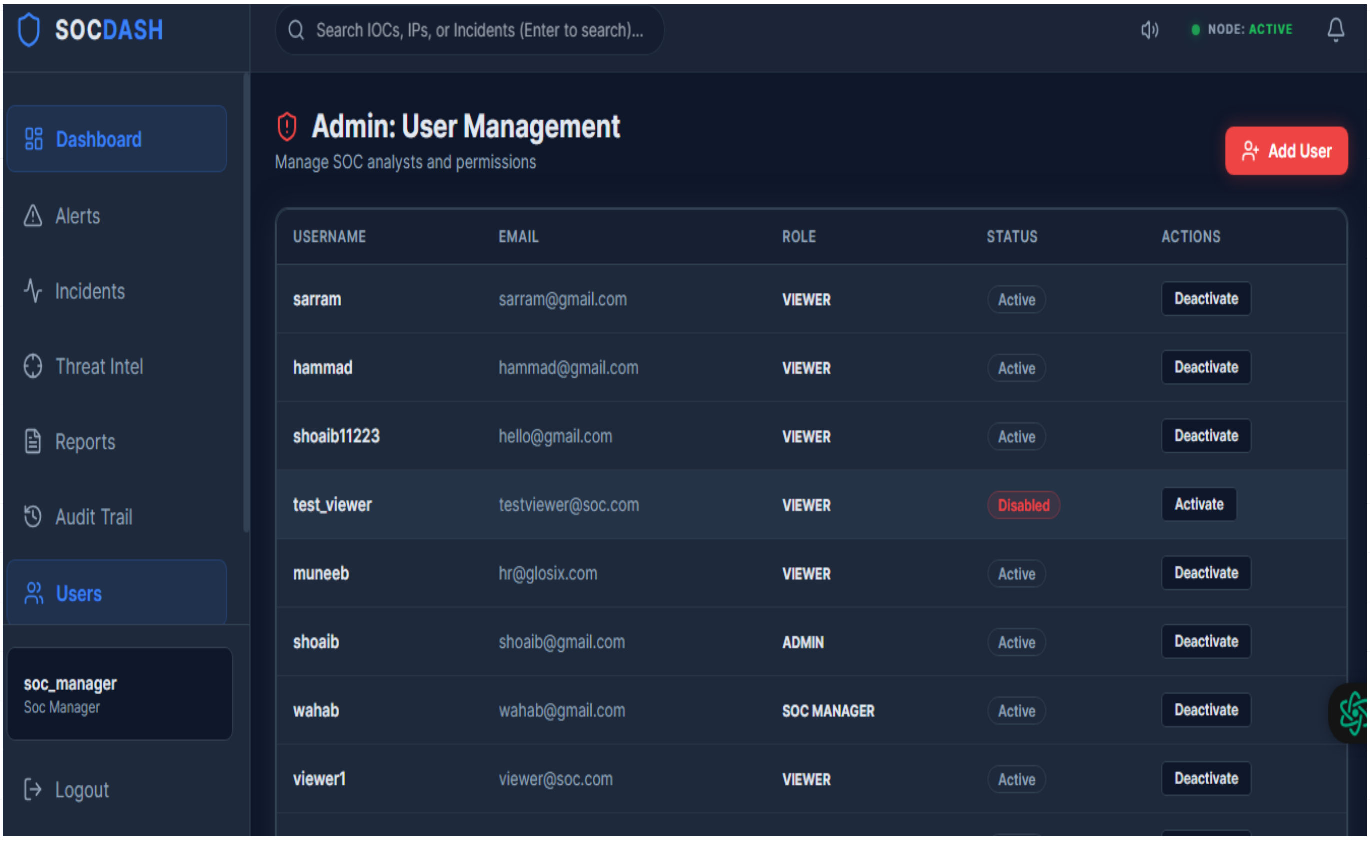
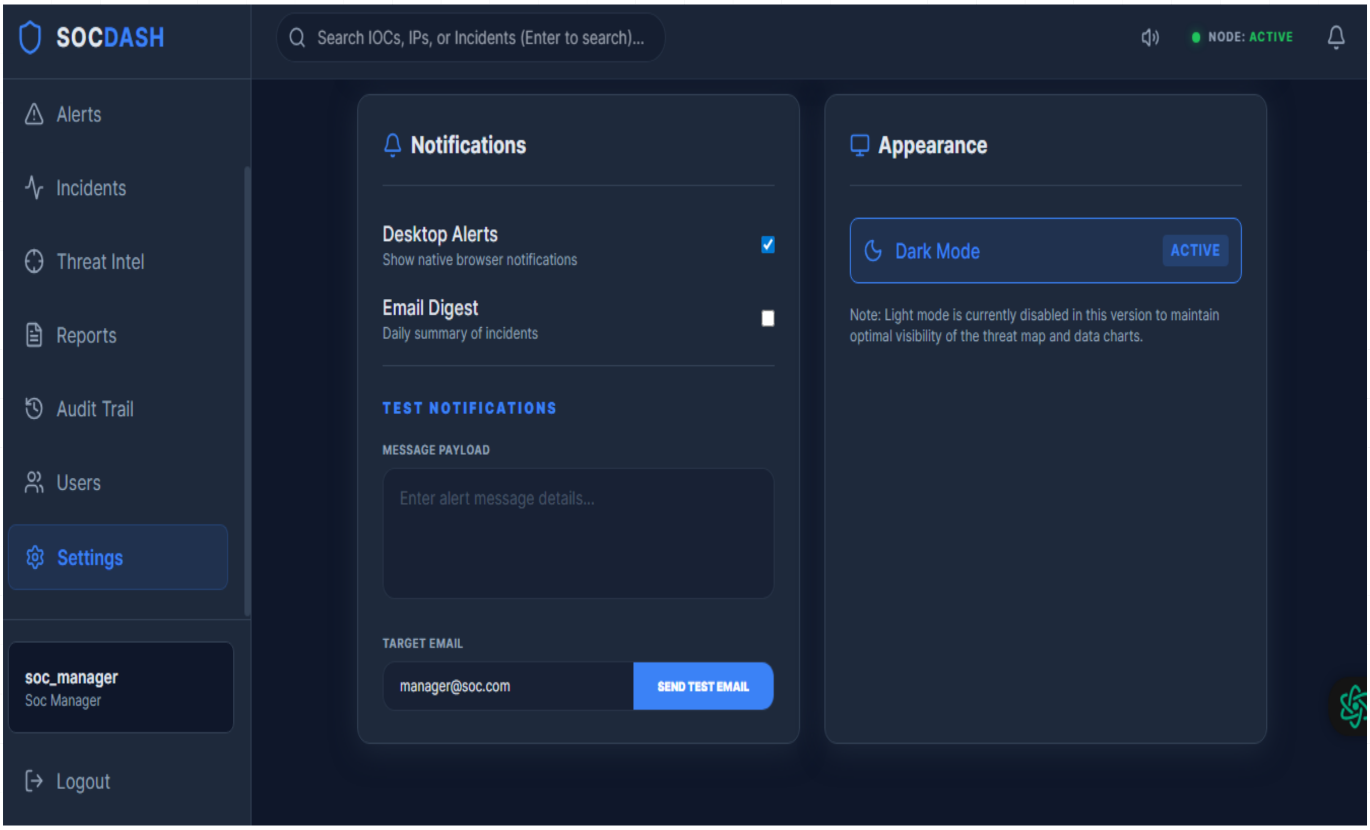


Figure 5.7 displays the user management interface for role-based access control.



System configuration options are shown in the settings page (Figure 5.8).

5.2 Performance Testing

Table 5.2: Performance Testing Result

Endpoint	Samples	Avg (ms)	Min (ms)	Max (ms)	Std Dev	Error %	Tput/s
Login	5	**217**	119	**553**	168.51	**0.00%**	0.104
Health Check	15	**9**	1	**98**	23.75	**0.00%**	0.312
Dashboard	15	**31**	7	**230**	54.19	**0.00%**	0.312
Get Alerts	15	**21**	8	**135**	30.51	**0.00%**	0.312
Alert Stats	15	**15**	8	**24**	5.27	**0.00%**	0.312
Incidents	15	**12**	6	**22**	4.86	**0.00%**	0.312
Create Alert	15	**8**	4	**16**	3.88	**0.00%**	0.312
TOTAL	**95**	**27 ms**	**1 ms**	553 ms	65.34	**0.00%**	**1.967**

5.3 Gantt Chart Overview

Table 5.3: Gantt Chart Overview

Phase	Timeline	Activities
Requirements & Research	Weeks 1	Gather requirements, conduct user research
UI/UX Design & Prototyping	Weeks 2–4	Create wireframes, design mockups, build prototypes
Backend & API Development	Weeks 4–6	Develop APIs, database schema, backend logic
Integration & Real-Time Features	Weeks 6–8	Implement WebSockets, integrate data sources
Security Testing & Penetration Testing	Weeks 8–11	Conduct security audits, vulnerability assessments
Deployment & Documentation	Weeks 11–13	Deploy to production, create user documentation
Presentation & Demo	Week 14	Final presentation and system demonstration

5.4 Conclusion

The SOC Dashboard project successfully delivers a modern, intuitive, and powerful platform for security operations. By integrating real-time data visualization, threat intelligence enrichment, and incident management into a unified interface, the system significantly enhances the efficiency and effectiveness of SOC teams.

The project was grounded in comprehensive user research, clearly defined requirements, and a scalable architecture designed for both SMB and enterprise environments. Security testing confirmed compliance with OWASP Top 10 guidelines, while performance testing validated the system's ability to handle production workloads.

The SOC Dashboard represents a practical, research-driven solution to real-world cybersecurity challenges, particularly within the context of Pakistan's growing need for accessible and effective security monitoring tools.

"The greatest risk when it comes to cyber security is ignorance."

— Kevin Mitnick